

Quiz (Habanero)

- Q1.** What is the solution to the system $x + y = 4$ and $x - y = 2$?
(A) $x = 3, y = 1$
(B) $x = 2, y = 2$
(C) $x = 1, y = 3$
(D) $x = 4, y = 0$
(E) $x = 0, y = 4$
- Q2.** The temperature in a greenhouse is modeled by $t = 2x + 5$ and $t = 3x - 2$. What is the intersection point?
(A) (1, 7)
(B) (2, 9)
(C) (3, 11)
(D) (4, 13)
(E) (5, 15)
- Q3.** The system $y = 2x - 1$ and $y = -3x + 4$ has a solution at
(A) (0, 0)
(B) (1, 1)
(C) (2, 3)
(D) (1, 1)
(E) (-1, 5)
- Q4.** The graphs of $y = x + 2$ and $y = 3x - 1$ intersect at
(A) (0, 0)
(B) (1, 1)
(C) (2, 4)
(D) (1, 3)
(E) (2, 5)
- Q5.** What is the solution to the system $x - 2y = 3$ and $2x + y = 5$?
(A) $x = 1, y = -1$
(B) $x = 2, y = 0$
(C) $x = 3, y = 1$
(D) $x = 4, y = 2$
(E) $x = 1, y = 2$
- Q6.** The amount of rainfall in a region is given by the system $r = 2t + 1$ and $r = t + 3$. What is the value of t when $r = 5$?
(A) $t = 1$
(B) $t = 2$
(C) $t = 3$
(D) $t = 4$
(E) $t = 5$
- Q7.** The system $y = x^2 - 2x - 3$ and $y = 2x - 1$ has a solution at
(A) (-1, -5)
(B) (0, 0)
(C) (1, 1)
(D) (2, 3)
(E) (3, 5)
- Q8.** The ecosystem of a forest is modeled by $p = 2n + 5$ and $p = n + 8$, where p is the population of a species and n is the number of years. What is the population when $n = 3$?
(A) $p = 10$
(B) $p = 11$
(C) $p = 12$
(D) $p = 13$
(E) $p = 14$

Open Ended Questions (FRQ)

- Q9.** Solve the system $x + 2y = 7$ and $3x - 2y = 5$ graphically and find the intersection point. Show your work.
- Q10.** The temperature in a city is given by the system $t = x + 2y$ and $t = 2x - 3y$, where x is the time of day and y is the day of the month. Find the temperature at (2, 1) and explain the solution graphically.

Chef's Tips

- Tip 1: Ensure students have graph paper for questions 9 and 10.
- Tip 2: Allow the use of calculators for checking calculations.
- Tip 3: Provide extra time for students who need accommodations.